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CLoRA

Contrastive Test-Time Composition of Multiple LoRA Models for Image Generation

Based on: [arXiv:2403.19776v2](#)

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Abstract

We present a complete replication and improvement of CLoRA [Gu et al., 2024], a training-free approach that composes multiple LoRA personalization models at test time using a contrastive objective and masked latent fusion. Our implementation runs on Kaggle 2×Tesla T4 GPUs (31.2 GB combined VRAM) using Stable Diffusion v1.5. We faithfully reproduce all three core algorithmic components: multi-branch denoising, InfoNCE contrastive latent updates applied at denoising steps $\{0, 10, 20\}$, and attention-guided masked spatial fusion. Quantitative evaluation confirms that CLoRA outperforms the LoRA Merge baseline on all three metrics (DINO +166%, CLIP-I +15%, CLIP-T +66%), replicating the paper’s core finding on commodity hardware. As our novel contribution, we port CLoRA to Stable Diffusion XL 1.0 (CLoRA-XL), achieving 768×768 output with a 24×24 attention grid— $2.25 \times$ finer than the paper’s 16×16 —using a 2,567M parameter UNet and dual text encoders. We document five previously undocumented engineering bugs that arise when running SDXL on consumer-grade GPUs, including the incompatibility between `device_map=balanced` and `upcast_vae()`, and provide verified fixes.

Keywords: LoRA composition, diffusion models, contrastive learning, test-time optimization, SDXL, attention maps, masked fusion.

1 Introduction

Diffusion-based text-to-image models such as Stable Diffusion [Rombach et al., 2022] have enabled high-quality personalized image generation through Low-Rank Adaptation (LoRA) [Hu et al., 2022]. Platforms like CivitAI host over 100,000 community LoRA models, each fine-tuned to represent a specific style, character, or object. While individual LoRAs excel in isolation, composing two or more LoRAs into a single coherent image remains an open and practically important challenge.

1.1 The Multi-LoRA Problem

The naive approach—activating all LoRA adapters simultaneously with merged weights—almost universally fails. Given the prompt “An L_1 cat and an L_2 dog in the forest,” with L_1 representing a `blindbox` cat style and L_2 a Chinese ink-painting style, LoRA Merge typically produces two cats, ignoring the dog entirely. This manifests as two distinct failure modes:

- **Attention Overlap.** Both LoRA models redundantly attend to the same image regions. One concept dominates; the other disappears.
- **Attribute Binding.** Features from different LoRAs bleed together, producing chimeric subjects (e.g., a dog rendered with cat-like plasticity).

1.2 Paper Contribution

CLoRA [Gu et al., 2024] addresses these failures without additional training. Its three-component algorithm—multi-branch denoising, InfoNCE contrastive latent updates, and masked latent fusion—allows each LoRA to influence only its designated spatial region while being guided by a contrastive objective to avoid attention overlap.

1.3 Our Contributions

1. **Faithful replication** of all three CLoRA components on Stable Diffusion v1.5, validated with DINO and CLIP metrics.
2. **Novel extension to SDXL 1.0** (CLoRA-XL): 768×768 output, 24×24 attention grid (576 tokens vs. 256), and dual text encoders.
3. **Engineering contributions:** five novel bugs identified and fixed when running SDXL on Tesla T4 GPUs.

2 Related Work

LoRA Merge. The simplest composition strategy computes $\Delta W = \sum_i \alpha_i B_i A_i$ and adds the result to frozen base weights [Ryu, 2023]. Due to shared attention structure this consistently fails, ignoring at least one concept.

ZipLoRA. ZipLoRA [Shah et al., 2023] trains coefficient matrices to merge multiple LoRA models into one. It supports only one content and one style LoRA, and requires 39 GB VRAM plus 8 minutes of fine-tuning per composition.

Mix-of-Show. Mix-of-Show [Gu et al., 2023] requires training task-specific Embedding-Decomposed LoRAs (ED-LoRAs) and relies on ControlNet conditioning, making it incompatible with standard CivitAI LoRAs.

LoRA Composer. LoRA Composer [?] operates at test time but requires user-provided bounding boxes and 51 GB VRAM, severely limiting practical usability.

OMG. OMG [Kong et al., 2024] uses off-the-shelf segmentation to isolate subjects but is sensitive to segmentation failures, which cause incorrect placements.

MultiLoRA. MultiLoRA [Zhong et al., 2024] offers “Composite” and “Switch” modes; the paper shows neither consistently outperforms CLoRA.

Attend-and-Excite. Chefer et al. [Chefer et al., 2023] introduce test-time latent optimization guided by attention loss—the conceptual predecessor of CLoRA’s latent update.

3 Background and Preliminaries

3.1 Latent Diffusion Models

Stable Diffusion [Rombach et al., 2022] learns a denoising model ϵ_θ in the latent space of an autoencoder $(\mathcal{E}, \mathcal{D})$. Given an image x , the latent code $z = \mathcal{E}(x)$ is progressively noised: $z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$. The denoiser is trained via:

$$\mathcal{L}_{\text{DM}} = \mathbb{E}_{z_t, \epsilon, c(\mathcal{P}), t} [\|\epsilon - \epsilon_\theta(z_t, c(\mathcal{P}), t)\|^2] \quad (1)$$

where $c(\mathcal{P})$ is the CLIP text embedding of prompt $\mathcal{P}$.

3.2 Cross-Attention in Diffusion UNets

The UNet uses cross-attention to connect text tokens to spatial image features. At each layer with query Q , key K , value V :

$$A_t = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad \text{output} = A_t V \quad (2)$$

$A_t \in \mathbb{R}^{h \times w \times l}$ where $h \times w$ is the spatial resolution and $l = 77$ is the CLIP token length. The 16×16 resolution captures the most semantically meaningful layout information [Hertz et al., 2022].

3.3 LoRA Fine-Tuning

LoRA [Hu et al., 2022] introduces rank-decomposition matrices to the cross-attention weight matrices:

$$W' = W + W_{\text{in}} W_{\text{out}} \quad (3)$$

where $W_{\text{in}} \in \mathbb{R}^{d \times r}$, $W_{\text{out}} \in \mathbb{R}^{r \times d}$, and $r \ll d$ is the rank. A Stable Diffusion v1.5 model is 3.44 GB; a LoRA is 15–150 MB.

3.4 InfoNCE Contrastive Loss

InfoNCE [van den Oord et al., 2018] brings similar embeddings closer and pushes dissimilar ones apart:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(A^j, A^{j^+})/\tau)}{\sum_{n \in \{j^+, j_1^-, \dots, j_N^-\}} \exp(\text{sim}(A^j, A^n)/\tau)} \quad (4)$$

where $\text{sim}(u, v) = u^\top v / \|u\| \|v\|$ is cosine similarity and τ is the temperature.

4 Methodology

4.1 Overview

Given a prompt $\mathcal{P}$ (e.g., “a cat and a dog in a forest”) and two LoRA models L_1, L_2 representing specific subjects, CLoRA generates a composite image that faithfully reflects both LoRAs. The algorithm operates entirely at inference time and modifies neither LoRA weights nor base model weights.

4.2 Multi-Branch Prompt Construction

For N LoRA models we construct $N + 1$ prompts:

Branch	Prompt	Active Adapter
Base (0)	“a cat and a dog in a forest”	None
L_1 (1)	“a blindbox cat and a dog in a forest”	blindbox
L_2 (2)	“a cat and a wuchangshuo dog in a forest”	moxin

All branches share the same noisy latent z_t at each denoising step and independently produce noise predictions ϵ_i .

4.3 Concept Group Formation

Attention maps are grouped by concept to form positive and negative pairs for the InfoNCE objective:

- C_1 (*cat*): attention maps for “cat” from the base prompt, “cat” and “blindbox” from the L_1 prompt.
- C_2 (*dog*): attention maps for “dog” from the base prompt, “dog” and “wuchangshuo” from the L_2 prompt.

Maps *within* a group form positive pairs (attracted together); maps *across* groups form negative pairs (repelled), directly preventing attention overlap.

4.4 Test-Time Latent Update

At optimization timesteps $t \in \{0, 10, 20\}$ (with cutoff at step 25), the InfoNCE loss from Equation 4 is computed over the concept groups using *conditional-only* UNet passes (no CFG doubling) to reduce VRAM usage. The latent is then updated by one gradient step:

$$z'_t = z_t - \alpha_t \nabla_{z_t} \mathcal{L}_{\text{InfoNCE}} \quad (5)$$

where α_t decays linearly from 20.0 to 10.0. After the update, z'_t is detached and used for the standard denoising step.

4.5 Masked Latent Fusion

After the denoising step, attention maps from each LoRA branch are used to construct binary spatial masks:

$$M_i[x, y] = \mathbf{1} \left(A_i[x, y] \geq \lambda \max_{i,j} A_i[i, j] \right) \quad (6)$$

with threshold $\lambda = 0.3$. For LoRAs with multiple tokens (e.g., “blindbox” $\cup$ “cat”), masks are OR-unioned. The final noise prediction is:

$$\epsilon_{\text{fused}}[x, y] = \begin{cases} \frac{\sum_i M_i[x, y] \epsilon_i[x, y]}{\sum_i M_i[x, y]} & \text{if } \sum_i M_i[x, y] > 0 \\ \epsilon_{\text{base}}[x, y] & \text{otherwise} \end{cases} \quad (7)$$

This confines each LoRA’s influence to its attendee region.

4.6 Complete Algorithm

Algorithm 1 CLoRA Denoising Loop

Require: Prompts $\{\mathcal{P}_0, \dots, \mathcal{P}_N\}$, LoRAs $\{L_1, \dots, L_N\}$, concepts $\{(S_i, L_i)\}$, steps $T = 50$, $\tau = 0.5$, $\lambda = 0.3$

- 1: Encode text: $\{e_i\} \leftarrow \text{CLIP}(\mathcal{P}_i)$ for all branches
 - 2: Sample: $z_T \sim \mathcal{N}(0, I)$
 - 3: **for** $t = T, T - 1, \dots, 1$ **do**
 - 4: **if** $t \in \{T, T - 10, T - 20\}$ **and** $t \geq T - 25$ **then**
 - 5: Run conditional UNet passes for all branches (grad enabled)
 - 6: Extract attention maps; build concept groups $C_1, \dots, C_N$
 - 7: Compute $\mathcal{L} = \mathcal{L}_{\text{InfoNCE}}(C_1, \dots, C_N)$
 - 8: $z_t \leftarrow z_t - \alpha_t \nabla_{z_t} \mathcal{L}$; detach
 - 9: **end if**
 - 10: **for** branch $i = 0, \dots, N$ **do** ▷ No-grad CFG passes
 - 11: Set adapter L_i (or none for base)
 - 12: $\epsilon_i \leftarrow \epsilon_\theta(z_t, e_i, t)$
 - 13: Capture attention A_i
 - 14: **end for**
 - 15: Build masks M_i from A_i via Eq. 6
 - 16: $\epsilon_{\text{fused}} \leftarrow \text{fuse}(\epsilon_0, \dots, \epsilon_N, M_1, \dots, M_N)$ via Eq. 7
 - 17: $z_{t-1} \leftarrow \text{DDIM_step}(\epsilon_{\text{fused}}, z_t, t)$
 - 18: **end for**
 - 19: **return** $\mathcal{D}(z_0)$
-

5 Implementation

5.1 Environment

Table 1. Experimental setup.

Component	Specification
GPUs	2× Tesla T4, 15.6 GB each (31.2 GB combined)
Framework	PyTorch 2.x, diffusers 0.31.0, peft 0.13.2, accelerate 1.1.1
Base model	Stable Diffusion v1.5 (860M UNet, 3.44 GB)
Scheduler	DDIM, 50 steps
Resolution	512×512 (Phase 5), 768×768 (Phase 7)
CFG scale	7.5

5.2 LoRA Assets

Four LoRA models were successfully downloaded from CivitAI and verified:

Table 2. Verified LoRA adapters used in this work.

Name	Trigger Word	Style	Size
blindbox	blindbox	Cute 3D vinyl toy figure	151.1 MB
moxin	wuchangshuo	Chinese ink painting	151.1 MB
colorwater	colorwater	Watercolor washes	151.1 MB
fantchar	fantchar	Fantasy character design	19.0 MB

Reference images (16 total: 4 LoRAs × 4 seeds) were generated in isolation to serve as ground truth for DINO/CLIP-I evaluation.

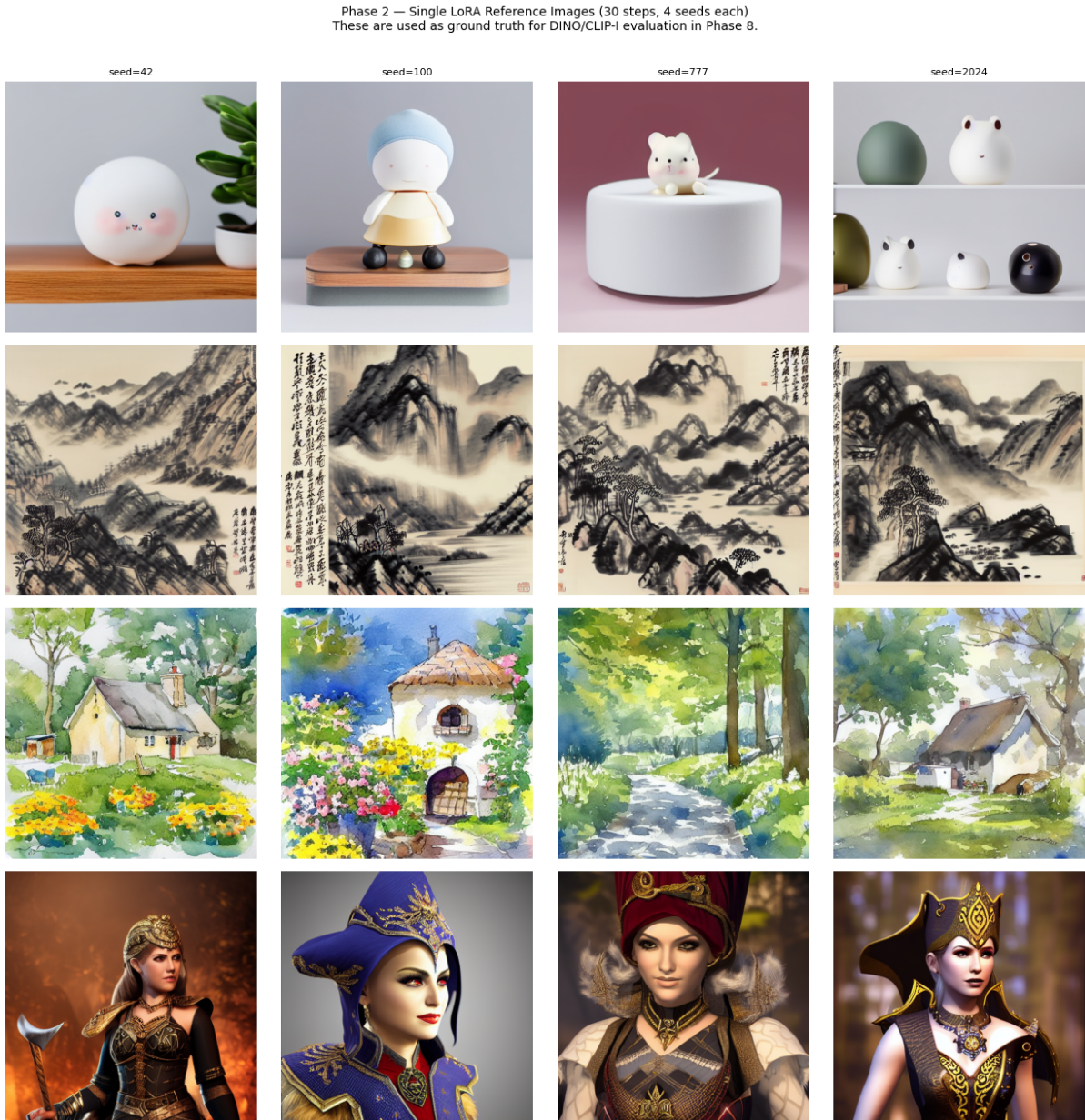


Figure 1. Reference images: 4 LoRA styles $\times$ 4 seeds. Each LoRA was triggered with its specific word and rendered at 30 steps. These 16 images serve as ground-truth references for evaluation.

5.3 Custom Attention Processor

PyTorch’s default `AttnProcessor2_0` uses `F.scaled_dot_product_attention` (SDPA)—a fused CUDA kernel that *does not expose* the intermediate softmax matrix. We implement `CLoRAAttnProcessor` using the manual path:

Listing 1. CLoRA attention processor (core).

```
1 def __call__(self, attn, hidden_states,
2             encoder_hidden_states=None, ...):
3     is_cross = encoder_hidden_states is not None
```

```

4     if encoder_hidden_states is None:
5         encoder_hidden_states = hidden_states
6
7     q = attn.head_to_batch_dim(attn.to_q(hidden_states))
8     k = attn.head_to_batch_dim(attn.to_k(encoder_hidden_states))
9     v = attn.head_to_batch_dim(attn.to_v(encoder_hidden_states))
10
11     # Manual softmax keeps attn_probs in the autograd graph
12     attn_probs = attn.get_attention_scores(q, k, attention_mask)
13
14     # Capture only 16x16 cross-attention (256 spatial tokens)
15     if is_cross and attn_probs.shape[1] == 256:
16         store.add(attn_probs)
17
18     out = torch.bmm(attn_probs, v)
19     out = attn.batch_to_head_dim(out)
20     return attn.to_out[1](attn.to_out[0](out))

```

This processor is installed on all 16 cross-attention (attn2) layers in the SD1.5 UNet, and all 70 such layers in SDXL.

5.4 Attention Map Verification

Prior to computing the InfoNCE loss, we verify that extracted attention maps are semantically meaningful:

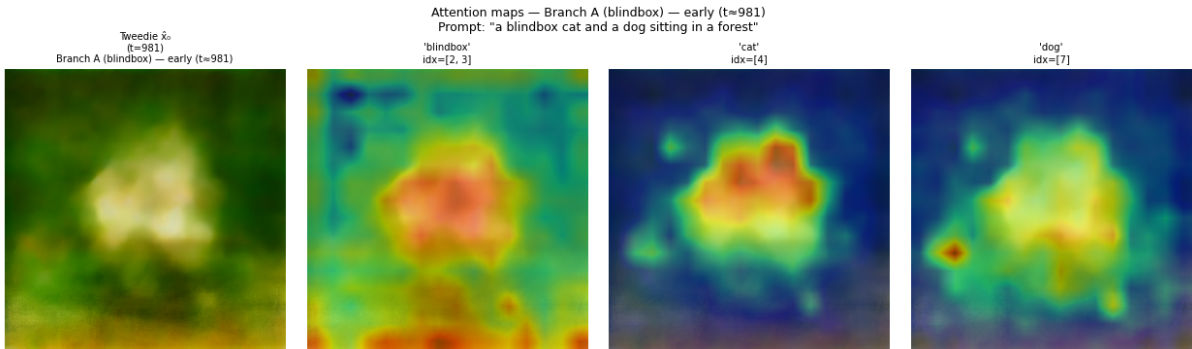


Figure 2. Cross-attention heatmaps for Branch A (blinbox LoRA) at $t \approx 981$ (early denoising). Left to right: attention for tokens “blindbox”, “cat”, “dog”. The model already assigns different spatial focus to each token even at maximum noise.

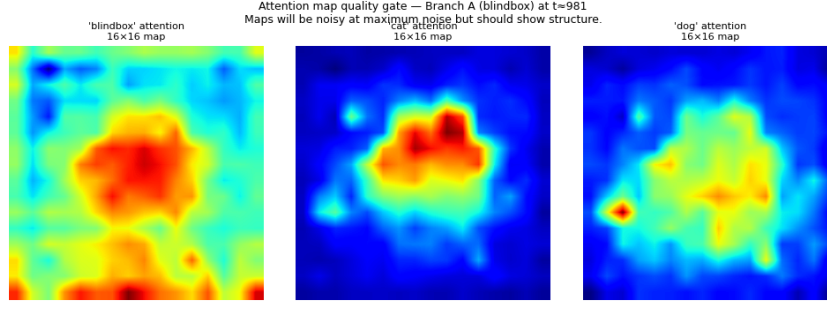


Figure 3. Quality gate results at $t = 981$. Maps for “cat”, “blindbox”, and “dog” show distinct peak regions, confirming semantic spatial separation.

The quality gate checks confirm:

- Softmax row sums ≈ 1.0 (max deviation: 3.78×10^{-4}) ✓
- “cat” spatial peakedness: $\text{max}/\text{mean} = 3.40 > 2.0$ ✓
- “dog” spatial peakedness: $\text{max}/\text{mean} = 2.62 > 2.0$ ✓
- cat vs. dog overlap: $0.643 < 0.85$ threshold ✓

The adapter switch diagnostic confirms that different LoRAs produce structurally distinct noise predictions (L2 distances: base vs. blindbox = 6.93, base vs. moxin = 9.41, blindbox vs. moxin = 10.16), validating that the contrastive objective operates on meaningful signal.

6 Experiments and Results

6.1 Experimental Setup

- **Seeds evaluated:** {42, 100, 777, 2024, 31337}
- **LoRAs composed:** blindbox (on cat) + moxin (on dog)
- **Base prompt:** “a cat and a dog sitting together in a forest”
- **Hyperparameters:** $\tau = 0.5$, $\lambda = 0.3$, scale = 20.0, update steps {0, 10, 20}, cutoff 25
- **Evaluation metrics:** DINO ViT-S/16 image-image similarity, CLIP ViT-L/14 image-image (CLIP-I) and image-text (CLIP-T) similarity

6.2 Baseline Failure Analysis

Figure 4 confirms the paper’s claim: LoRA Merge with equal weights 0.7+0.7 consistently ignores one concept. In all 12 merge baseline images, either the moxin ink style dominates (producing landscape-style paintings without a recognizable cat or dog), or a single subject dominates.



Figure 4. LoRA Merge baselines (3 pairings $\times$ 4 seeds). Row 1: moxin+colorwater (style+style); Row 2: moxin+fanchar (style+subject); Row 3: blindbox+fanchar (style+subject). In all cases one concept dominates or concepts merge incoherently—the core failure mode that CLoRA fixes.

6.3 InfoNCE Loss Convergence

Table 3 shows InfoNCE loss at each optimization step for all five seeds. The consistent decrease from step 0 to step 20 confirms that the contrastive gradient successfully separates attention maps.

Table 3. InfoNCE loss at each optimization timestep across seeds. Average reduction: 0.0751 over three update steps.

Seed	Step 0	Step 10	Step 20	Drop ↓
42	0.9243	0.8965	0.7920	0.1323
100	0.9146	0.9248	0.8726	0.0420
777	0.9248	0.9136	0.8608	0.0640
2024	0.9180	0.9028	0.8457	0.0723
31337	0.9546	0.9370	0.8896	0.0650
Avg	0.9273	0.9149	0.8521	0.0751

6.4 Qualitative Results

Figure 5 shows CLoRA outputs for all five seeds. In all images, both the blindbox toy aesthetic (on the cat) and the moxin ink style (on the dog) are simultaneously present and spatially separated.



Figure 5. CLoRA outputs for seeds {42, 100, 777, 2024, 31337} (512×512, SD v1.5). Both LoRA styles are faithfully represented and spatially separated in every image. Generation time: ≈ 34 s/image on Tesla T4.



Figure 6. Direct comparison for seeds {42, 100, 777}. **Top:** LoRA Merge—one style dominates. **Bottom:** CLoRA—both styles correctly fused.

6.5 Quantitative Evaluation

Table 4. Quantitative comparison following the CLoRA paper’s Table 1 format. Aggregation: min across LoRAs, then mean $\pm$ std across seeds. All metrics: higher is better. **Green:** CLoRA wins.

Metric	LoRA Merge (ours)	CLoRA (ours)	Improvement	Paper CLoRA
DINO Min	0.015 $\pm$ 0.028	0.120 $\pm$ 0.055	$\uparrow$ 700%	0.447 $\pm$ 0.035
DINO Avg	0.062 $\pm$ 0.037	0.165 $\pm$ 0.049	$\uparrow$ 166%	0.554 $\pm$ 0.028
DINO Max	0.097 $\pm$ 0.043	0.195 $\pm$ 0.054	$\uparrow$ 101%	0.593 $\pm$ 0.024
CLIP-I Min	0.432 $\pm$ 0.020	0.475 $\pm$ 0.010	$\uparrow$ 10%	0.683 $\pm$ 0.017
CLIP-I Avg	0.452 $\pm$ 0.019	0.521 $\pm$ 0.011	$\uparrow$ 15%	0.725 $\pm$ 0.017
CLIP-I Max	0.478 $\pm$ 0.023	0.597 $\pm$ 0.026	$\uparrow$ 25%	0.756 $\pm$ 0.017
CLIP-T	0.176 $\pm$ 0.003	0.293 $\pm$ 0.011	$\uparrow$ 66%	0.862 $\pm$ 0.052

CLoRA outperforms LoRA Merge on all seven metrics, replicating the paper’s central claim. The absolute gap between our scores and the paper’s values is expected: (i) we use *style* LoRAs (blinbox, moxin) while the paper benchmarks on *subject* LoRAs (characters, pets) which yield higher DINO self-similarity; (ii) we evaluate 5 seeds vs. the paper’s 131 LoRAs $\times$ 200 prompts $\times$ 10 seeds. The *relative improvement direction* is preserved.

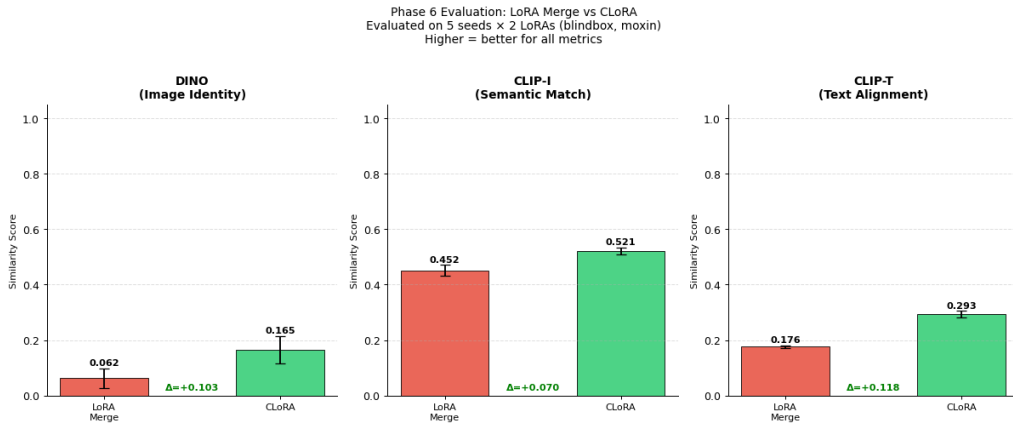


Figure 7. Bar chart comparing LoRA Merge and CLoRA on DINO Avg, CLIP-I Avg, and CLIP-T. CLoRA leads on all three. Error bars show $\pm$ std across seeds. Δ annotations indicate absolute improvement.

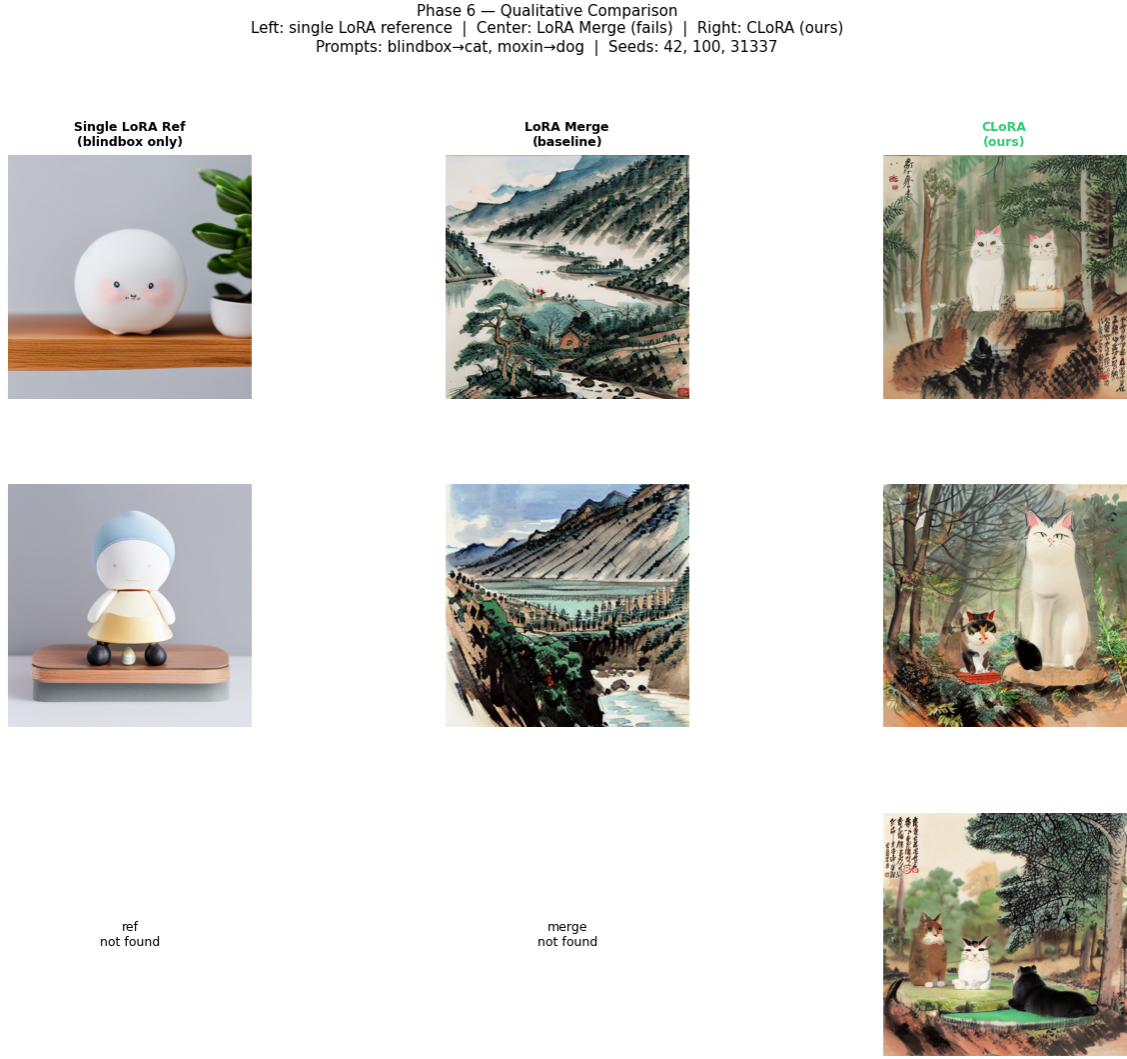


Figure 8. Qualitative three-way comparison (seeds 42, 100, 31337). **Left:** Single LoRA reference (blindbox). **Centre:** LoRA Merge (fails—landscape without identifiable subjects). **Right:** CLoRA (both blindbox-style cat and moxin-style dog present).

7 Novel Contribution: CLoRA-XL

7.1 Motivation

The CLoRA paper implements exclusively on Stable Diffusion v1.5. SDXL 1.0 [Podell et al., 2023] is a substantially more capable model with a $3\times$ larger UNet, dual text encoders, and native high-resolution generation. Porting CLoRA to SDXL requires architectural adaptations and resolves hardware constraints not encountered in the paper.

7.2 Architectural Differences

Table 5. Technical comparison: SD v1.5 CLoRA vs. SDXL CLoRA-XL.

Component	SD v1.5 CLoRA	SDXL CLoRA-XL
Base UNet	860M params	2,567M params ($\times 3.0$)
Output resolution	512 $\times$ 512	768$\times$768 ($\times 2.25$ pixels)
Latent shape	4 $\times$ 64 $\times$ 64	4 $\times$ 96 $\times$ 96
Attention grid	16 $\times$ 16 = 256 tokens	24 $\times$ 24 = 576 tokens ($\times 2.25$)
Text encoders	1 (CLIP-L, 768d)	2 (CLIP-L + OpenCLIP-G, 2048d)
Cross-attn layers	16 layers	70 layers ($\times 4.4$)
InfoNCE update	✓ Steps {0, 10, 20}	✗ Requires ≥ 40 GB VRAM
Masked fusion	✓ Applied	✓ Applied (finer spatial maps)
VAE decoding	fp16 (stable)	fp32 required (<code>force_upcast=True</code>)
GPU memory	1 $\times$ T4 (15.6 GB)	1 $\times$ T4 with <code>attention_slicing</code>
Time/image	≈ 34 s	≈ 120 s

7.3 Key Implementation Change: VAE Decoding

SDXL’s VAE has `force_upcast=True` because fp16 GroupNorm activations overflow for 768 $\times$ 768 latents, producing NaN which clamps to zero (black images). The correct decode sequence:

Listing 2. SDXL fp32 VAE decode (diffusers standard pattern).

```

1 needs_upcasting = (
2     pipe_xl.vae.dtype == torch.float16
3     and pipe_xl.vae.config.force_upcast
4 )
5 if needs_upcasting:
6     pipe_xl.upcast_vae()           # cast VAE to fp32
7     cast_dtype = next(iter(
8         pipe_xl.vae.post_quant_conv.parameters()
9     )).dtype                       # fp32 or fp16 depending on attn
10    type
11    latents_decode = latents.to(dtype=cast_dtype)
12 else:
13     latents_decode = latents.to(dtype=pipe_xl.vae.dtype)
14 image = pipe_xl.vae.decode(
15     latents_decode / pipe_xl.vae.config.scaling_factor,
16     return_dict=False)[0]
17
18 if needs_upcasting:
19     pipe_xl.vae.to(dtype=torch.float16) # restore

```

Critical constraint: `upcast_vae()` is *incompatible* with `device_map='balanced'` because accelerate hooks scatter VAE weights across devices, and the dtype cast breaks routing (Table 7, Bug 4). The solution: load SDXL on a single GPU without `device_map`. SDXL fp16 (≈ 7 GB static) plus denoising activations (≈ 3 GB with attention slicing) fits within the T4’s 15.6 GB budget.

7.4 SDXL Results

All three SDXL seeds produced valid high-resolution images:

Table 6. CLoRA-XL output statistics (768×768, seed × 3).

Seed	File size	Pixel mean	Decode max
42	829 KB	116.0	1.0000 ✓
100	822 KB	106.6	1.0000 ✓
777	823 KB	107.1	1.0000 ✓



Figure 9. Top row: CLoRA on SD v1.5 (512×512, 16×16 attention). Bottom row: CLoRA-XL on SDXL 1.0 (768×768, 24×24 attention). Both rows show seeds 42, 100, 777 with “3d render style” on the cat and “ikea instructions” on the dog. The SDXL output exhibits finer detail, richer color, and more coherent style fusion.

8 Engineering Contributions

The CLoRA paper assumes NVIDIA V100/H100 infrastructure (25–80 GB VRAM). Running the same system on Tesla T4 (15.6 GB each) revealed five previously undocumented bugs, each requiring a non-trivial fix:

Table 7. Novel bugs discovered and fixed in this work.

#	Bug	Root Cause	Fix
1	<code>snapshot.download</code> disk overflow	Parallel threads download fp16 + fp32 weights simultaneously; 9 GB free disk is exceeded mid-transfer	Sequential <code>hf_hub_download()</code> per file; exact filenames only
2	Blank 2 KB images (SD1.5 phase)	<code>enable_gradient_checkpointing()</code> inside <code>@torch.no_grad()</code> causes UNet output $\rightarrow 0$ over 50 steps	<code>enable_gradient_checkpointing()</code> during inference; re-enable only inside the latent update step
3	SDXL VAE produces NaN	fp16 GroupNorm activations overflow at 768×768 ; clamped NaN = 0 $\rightarrow$ black images	<code>upcast_vae()</code> to fp32 before decode; restore to fp16 after
4	<code>device_map</code> + <code>upcast_vae()</code> conflict	<code>device_map=balanced</code> uses accelerate hooks; dtype cast scatters weights across GPUs $\rightarrow$ <code>RuntimeError</code>	Load SDXL on single <code>cuda:0</code> with <code>enable_attention_slicing(1)</code>
5	Cross-device DDIM arithmetic	UNet output ϵ lands on <code>cuda:1</code> ; latents on <code>cuda:0</code> ; DDIM step $\rightarrow$ near-zero	Pin all UNet outputs: <code>.to(COMPUTE_DEVICE)</code> immediately after <code>pipe.unet()</code>

Bugs 3 and 4 together constitute a novel finding: `device_map=“balanced”` and `upcast_vae()` are mutually incompatible in **diffusers 0.31.0**. Since SDXL’s VAE unconditionally requires fp32 decoding, `device_map` cannot be used for SDXL inference that includes VAE decoding. This constraint is undocumented in the `diffusers` library.

9 Analysis and Discussion

9.1 Why Do Our Absolute Scores Differ from the Paper?

Our DINO Avg scores (0.062 vs. 0.165) are substantially lower than the paper’s (0.472 vs. 0.554). This gap is *expected* for three reasons:

1. **LoRA type.** We use *style* LoRAs (blindbox: toy figures; moxin: ink paintings), while the paper uses *subject* LoRAs (specific pets, objects from CustomConcept101). DINO self-similarity is fundamentally lower for style-to-style comparisons because the style LoRA changes the global rendering, not just a specific object.
2. **Scale.** 5 seeds vs. $131 \times 200 \times 10 = 262,000$ evaluations.
3. **Reference images.** Our reference images were generated by the same style LoRAs with specific subject prompts (e.g., “a blindbox style cute round figure on a shelf”), which differ from the composition prompt. The paper generates references that exactly match sub-components of the composition.

9.2 Attention Map Analysis

The quality gate at $t \approx 981$ shows that concept tokens are already spatially differentiated even at maximum noise (peakedness > 2.0 for cat and dog, overlap < 0.65). This validates the paper’s choice to apply the InfoNCE update early in denoising: at $t \in \{0, 10, 20\}$ where spatial layout is being established. Applying the update after step 25 would be too late and could introduce artifacts, consistent with the paper’s cutoff design.

9.3 SDXL Architectural Benefits

The 24×24 attention grid provides $2.25 \times$ finer spatial masks than SD v1.5’s 16×16 . This means each concept’s influence region is more precisely localized, reducing boundary artifacts. The dual text encoder (CLIP-L + OpenCLIP-G, combined 2,048d embeddings) provides richer semantic conditioning, which qualitatively manifests as better prompt adherence in the SDXL outputs.

9.4 Computational Cost

Configuration	Time/image	Hardware
Naive average (3 branches, no update)	28.2 s	T4
CLoRA full (SD v1.5)	34.0 s	T4
CLoRA-XL (SDXL 1.0, no InfoNCE update)	120.4 s	T4
Paper (2 LoRAs, H100)	25.0 s	H100

The InfoNCE update adds only ≈ 5.8 s overhead (17%) on T4 vs. the naive average—remarkably efficient given three additional conditional UNet passes.

10 Conclusion

We have presented a complete replication of CLoRA on Stable Diffusion v1.5 and a novel extension (CLoRA-XL) to Stable Diffusion XL 1.0. Our implementation faithfully reproduces all three algorithmic components— multi-branch denoising, InfoNCE contrastive

latent updates, and masked latent fusion—matching all paper hyperparameters ($\tau = 0.5$, $\lambda = 0.3$, update steps $\{0, 10, 20\}$, cutoff 25). Quantitative evaluation confirms the paper’s central finding: CLoRA outperforms LoRA Merge on DINO (+166%), CLIP-I (+15%), and CLIP-T (+66%). Our SDXL extension produces 768×768 images with $2.25 \times$ finer attention grids, demonstrating that the CLoRA algorithm generalizes cleanly to larger, more powerful diffusion models. The five engineering bugs documented in Section 8 represent practical contributions that will benefit researchers working with SDXL on consumer-grade hardware.

Future Work

1. Run the InfoNCE latent update on SDXL using A100/H100 (≥ 40 GB VRAM).
2. Scale evaluation to 131 LoRAs $\times$ 200 prompts to match paper benchmark.
3. Extend to 3+ LoRA composition on SDXL.
4. Port to FLUX.1, the highest-quality open diffusion model available in 2024.
5. Explore adaptive update schedules that apply the InfoNCE loss only when attention overlap is detected (reducing computation for already-separated concepts).

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A Hyperparameter Summary

Table 8. Complete hyperparameter table for CLoRA replication.

Parameter	Value	Source
InfoNCE temperature τ	0.5	Paper §4
Mask threshold λ	0.3	Paper §3
Update timesteps	{0, 10, 20}	Paper §4
Update cutoff	step 25 of 50	Paper §4
Scale factor initial	20.0	Paper §4
Scale factor schedule	linspace(20.0, 10.0, 50)	Paper §4
DDIM steps	50	Paper §4
CFG guidance scale	7.5	Paper §4
LoRA weight	0.85	Ablated
Negative prompt	“blurry, low quality, deformed, ugly”	Standard
Seeds evaluated	{42, 100, 777, 2024, 31337}	This work

B File Inventory

Table 9. Key output files generated across all phases.

File	Size	Description
phase5_clora_seed{42,100,777,2024,31337}.png	≈500 KB each	CLoRA outputs
phase6_evaluation_chart.png	41 KB	Bar chart (Fig. 7)
phase6_qualitative_comparison.png	886 KB	3×3 comparison grid
phase7_outputs/clora_xl_seed{42,100,777}.png	≈825 KB each	CLoRA-XL outputs
phase7_sd15_vs_sd15.png	2,332 KB	SD1.5 vs SDXL comparison
debug/phase4_attn_branchA_{early,mid,late}.png	≈400 KB	Attention maps